



The impact of movie titles on box office success[☆]

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ABSTRACT

The movie lifecycle comprises production, distribution, and exhibition factors. In general, the most significant expenses are associated with the production factors. However, the movie title is one of the relatively inexpensive production factors that a studio can use to its advantage. This study examines how an informative movie title, that is, a movie title that contains information about its genre or storyline, affects a movie's box office revenue. This study used an analysis of a 5-year period in the South Korean movie market to show that an informative movie title has a positive impact on box office revenue for an under-promoted movie, where prerelease promotional activities are measured by the volume of prerelease media exposure. However, this impact decreases as prerelease promotional activities increase. This mechanism not only contributes to the understanding of the film industry, but also to the literature on the entertainment industry and brand and firm names.

1. Introduction

Contrary to common belief, a high production budget has not been a necessary condition for box office success in the history of filmmaking. The movie *Night of the Living Dead* released in 1968 earned \$30 million worldwide, 263 times the production budget of \$114,000. In 2009, *Paranormal Activity*, which had a \$450,000 budget, earned \$194 million worldwide. *My Big Fat Greek Wedding*, which was released in 2002, earned \$374 million worldwide on a \$5 million budget and is a well-known example of a “sleeper hit,” a movie that achieves unexpected success without heavy promotion (e.g., Dellarocas, Zhang, & Awad, 2007; Karniouchina, 2011). However, as shown by numerous studies in the literature, it cannot be denied that the budget is one of the major factors that determines box office success (e.g. Basuroy, Chatterjee, & Ravid, 2003).

Movies that have a large budget are usually associated with a famous director, well-known cast, and generous marketing budget with which promotional efforts are focused immediately before the opening of the movie. Movie trailers appearing on television, online, and in theaters; press junkets featuring the stars and creators of the movie; and various publicity events all make the movie accessible in moviegoers' minds and help maintain people's awareness of it when they are considering which movies to see in the future (e.g., Zufryden, 2000). A countless number of low-budget films have disappeared without ever reaching consumer awareness due to meager promotional activities. As

a rule of thumb, the promotional budget takes up about 50% of the total production budget (Vogel, 2001), so it is highly likely that promotion for movies with a low production budget cannot reach the tipping point of influencing consumer choice.

With this in mind, the producers or studios making low-budget movies should turn to the most cost-effective ways to market them. One tool that has been underexplored is the title, or the naming of the movie. A movie title is one of the first pieces of information delivered to potential viewers, and it can provide information about the movie or even evoke curiosity, encouraging potential viewers to seek more information and ultimately affecting consumer choice. In the aforementioned examples, *Night of the Living Dead*, *Paranormal Activity*, and *My Big Fat Greek Wedding* were titles that made it easy for moviegoers to interpret the genre or storyline of the movie. Moreover, the title requires minimal to no cost to determine.

However, despite this unique characteristic of movie titles, as far as we know, almost no study in academia has examined the impact of titles on market performance, let alone in the entertainment industry, in contrast to broad discussions about the importance and effects of brand names and firm names in the general marketing literature (e.g., Horsky & Swyngedouw, 1987; Kalaignanam & Bahadir, 2013; Peterson & Ross, 1972; Rao & Monroe, 1989). To address this gap in the literature, this study examines how an informative movie title, that is, a movie title that contains information about its genre or storyline, affects its box office performance. Notably, the study finds that for under-promoted

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Table 1
Factors explaining box office revenue of movies.

Stakeholder factors (under control)		Third-party factors (not under control)
Production	- Genre: Dellarocas et al. (2007); Sawhney and Eliashberg (1996); Zhang, Luo, and Yang (2009) - Star power: Sawhney and Eliashberg (1996) - Sequel: Sood and Drèze (2006) - Country: Karniouchina (2011); Zhang et al. (2009) - Movie title: <i>Current Study</i>	- Media activity: Mestyán, Yasseri, and Kertész (2013)
Distribution	- Advertising Expenditure: Prag and Casavant (1994) - Internet promotion: Zufryden (2000) - MPAA ratings: Joshi and Hanssens (2009); Ravid (1999)	- Prerelease e-WOM: Liu (2006) - Critic review: Elberse and Eliashberg (2003); Eliashberg and Shugan (1997); Joshi and Hanssens (2009) - Media activity: Mestyán et al. (2013)
Exhibition	- Screen: Elberse and Eliashberg (2003); Karniouchina (2011); Neelamegham and Chintagunta (1999); Reinstein and Snyder (2005) - Revenue of first week: Sawhney and Eliashberg (1996) - Release date: Elberse and Eliashberg (2003); Ghiassi, Lio, and Moon (2015); Joshi and Hanssens (2009); Zhang et al. (2009)	- Postrelease e-WOM: Chintagunta, Gopinath, and Venkataraman (2010); Dellarocas et al. (2007); Duan, Gu, and Whinston (2008); Karniouchina (2011); Liu (2006) - Competition: Ainslie, Drèze, and Zufryden (2005) - Award: Nelson et al. (2001)

movies, box office revenues increase when the movie titles are informative. However, as the level of promotion increases, the impact of titles on revenue decreases because the consumers become more aware of the movie's genre or storyline from other sources, such as the media. The next section introduces literature on factors affecting movie performance, and the influence of names on the market outcomes for brands, firms, and movies. Then, we propose the research hypotheses, explain the data and methodology, present and discuss the results, and provide conclusions about the implications of the research.

2. Theoretical background and hypotheses

2.1. Factors influencing box office performance

The motion picture industry has gained much attention in the marketing literature over the past few decades, mostly owing to its cultural significance and impact on the global economy ([Eliashberg, Elberse, & Leenders, 2006](#)). In its lifecycle, a movie goes through production, distribution, and exhibition stages.¹ The movie lifecycle is often considered as a value chain because capital is injected or collected at each stage. In the value chain, studios, distributors, and exhibitors, which we refer to as stakeholders in this study, control various factors that affect the market performance of a movie. Production is the film-making stage, which includes the whole process of idea generation, screenplay writing, financing, arranging for the cast and crew, actual filming, and film editing and addition of various effects. Distribution consists of the physical distribution of the movies to theaters, such as timing of release and release format decisions (i.e., wide release or limited release) and prerelease marketing activities. Prerelease promotion is considered important for creating excitement (“buzz”) about a movie before it opens and generally includes trailers, advertising campaigns, press junkets, and interviews with the cast and directors. Finally, exhibition corresponds to presenting the movies for the viewers to watch. During this stage, exhibitors analyze the responses of initial viewers and adjust the number of allocated screens. For a detailed description of each stage of the value chain, see [Eliashberg et al. \(2006\)](#).

In addition to the above factors, third parties, such as general viewers and critics, can also affect a movie's box office revenue in the value chain. Widely discussed third-party factors include electronic word-of-mouth (e-WOM) and critic reviews. These are considered as more credible sources of information by viewers because they are less prone to control by producers and distributors than traditional promotional tools, such as advertising, and as such they have received

more attention in the literature recently (e.g., [Dellarocas et al., 2007](#); [Liu, 2006](#)). E-WOM, reviews, and opinions of movies generated by general viewers is a non-expert information source and primarily influences viewer choice, and ultimately box office revenue, during the distribution (prerelease e-WOM) and exhibition stages (postrelease e-WOM) ([Liu, 2006](#)). Critic reviews, which are an expert information source, are written by critics who make a living by reviewing movies and are mainly published in media outlets during the distribution stage ([Eliashberg & Shugan, 1997](#)). Other third-party factors that affect movie revenue at the exhibition stage include nominations and awards ([Nelson, Donihue, Waldman, & Wheaton, 2001](#)).

Thus, factors that explain a movie's revenue can be divided into stakeholder-controlled and third-party driven factors, each of which can have an influence throughout the movie value chain. [Table 1](#) presents the factors and related studies. The topic of interest in this study, that is, the movie title, is a stakeholder-driven factor in the production stage.

As shown in [Table 1](#), many stakeholder factors require high costs to determine and are difficult to change once determined. For example, the genre, screenplay, or actors are difficult to change once production (especially the filming process) begins, due to the time and resources already invested. Further, promotional expenses and screen allocations are costly factors that take a hefty amount of the total budget of the movie. However, the movie title is a stakeholder component that can easily be determined at a relatively low cost. Therefore, exploring the impact of movie titles will help provide stakeholders with insights to enhance their management of the performance of their movies. In the following section, we examine the general literature on the impact of product or brand names on market outcomes.

2.2. Impact of brand, firm, and movie names on market outcomes

During the product purchase process, consumers often have uncertainty about the product's quality or whether it is well matched to their goal/preference. This uncertainty is one of the major factors of a consumer's perceived risk in the buying decision process ([Cox & Rich, 1964](#)). A brand name is an attribute of the product that consumers are exposed to early in the information search stage and enables consumers to infer various expectations about the product's quality or whether the product can fulfill their goal/preference. Furthermore, by merely being exposed to the brand name, the consumer may or may not decide to further pursue an information search on this brand. Thus, a brand name reduces the search cost and cognitive effort when consumers evaluate products ([Landes & Posner, 1987](#)) and reduces the perceived risk of product quality ([Moon & Millison, 2000](#)). However, consumer choice based on brand name was found to not always be a rational choice. For example, in the fund market, [Cooper, Gulen, and Rau \(2005\)](#) found that

¹ In this study we focus our attention on theatrical motion pictures, movies that are released primarily for viewing in theaters, then released in other forms, such as internet streaming and DVD/Blu-ray.

the flow of funds increased when a mutual fund changed its name to the currently “hot” investment style, even if the fund did not change its portfolio, implying that names play an important role in decision making. Because of the importance of names, marketers and researchers have investigated various aspects of product or brand naming.

A “good” brand name increases brand awareness and positively affects product evaluation and purchase decisions (Aaker & Keller, 1990). The product's brand name is used by consumers as a clue about its quality and they extrapolate the quality of an unknown product indirectly through its brand name. In other words, an unknown product's brand compensates for the lack of information about the product. In a study of online shopping sites, Degeratu, Rangaswamy, and Wu (2000) showed that the brand name increasingly becomes more important as information about the attributes of the product decreases. The name effect exists even in cases where there is no prior awareness or knowledge of a brand. Peterson and Ross (1972) demonstrated how consumers understand a product through analogies of its brand name. They asked participants whether computer-generated words were appropriate for brand names depending on product type, and the participants showed a preference for specific names for particular product categories. Moreover, studies on brands found out that brand names can create consumer prejudice (Leclerc, Schmitt, & Dubé, 1994; Pavia & Costa, 1993; Zinkhan & Martin Jr., 1987), and the linguistic characteristics of brand names influence market outcomes (Grewal, Krishnan, Baker, & Borin, 1998; Heath, Chatterjee, & France, 1990).

Further, several studies have investigated the influence of firm names on investor behavior. Kalaianam and Bahadir (2013) found that a public announcement of a firm's name change that occurs at the same time as a restructuring gives investors much more information than if they occur at separate times. Changing the name of a firm was also found to relieve uncertainty about the firm. Horsky and Swyngedouw (1987) showed that changing a firm's name can increase its performance and profits, while Mase (2009) showed that changing a firm's name increases its value in a study of UK companies.

So far, the impact of brand and firm names on market outcomes has been examined. Surprisingly, however, almost no study has explored the impact of movie titles in the entertainment domain. One exception is Sood and Drèze (2006), who showed that a named sequel (e.g., *Daredevil: Taking It to the Streets*) resulted in higher user ratings than a numbered sequel (e.g., *Daredevil 2*), where “sequel” is defined as corresponding to a brand extension. However, their study differs from this study in that they only covered the naming of sequels, whereas we examine the role of informative titles in general.

2.3. Impact of informative movie titles on box office revenue for under-promoted movies

The mechanism of movie titles influencing consumer choice is expected to be analogous to that of brand and firm names; that is, movie titles are expected to reduce consumers' search costs and lower the perceived risk. Consumers do not analyze all the available data to create a consideration set when they choose a movie. Because a tradeoff occurs between the information-processing cost and the benefit from the choice, some items are not even processed at all (Yang, Toubia, & De Jong, 2015). Therefore, consumers obtain information about only a few movies when choosing a movie to watch. For a heavily promoted movie, such as a blockbuster, consumers are more likely to be aware of information about the movie's genre, storyline, synopsis, and cast. However, movies with a lower promotional budget cannot be promoted to the same extent as movies with a large budget. Thus, delivering information about such movies to consumers is more difficult. Moreover, because of the limited promotional budget, distributors of under-promoted movies are sometimes not able to deliver any information to consumers except for the titles. Consequently, for under-promoted movies, the likelihood that they will be chosen by consumers increases when their titles are informative, that is, if information such as genre or

Table 2

Examples of coding of Information variable.

Movie titles where Information = 0	Movie titles where Information = 1
Joe	Storks, a Baby Carrier
A Company Man	If I Ask Her Out
Broken	Eye to See the Ghost: Child's Eye
Condole Flies.	Death Of A Samurai
Lucy	Dragon Hunters

Note: the movies titles were originally in Korean and translated in English.

storyline can be inferred from the title. If consumers are intrigued by these initial details, they may seek out more information about the movie and eventually choose to watch it. Hence, we hypothesize that an informative movie title positively affects box office revenue.

H1. An informative movie title positively affects box office revenue.

2.4. Moderating effect of prerelease media coverage

However, this hypothesis is limited to cases where consumers do not have much prior information about the movie in question. As the degree of promotion of the movie increases, consumers can easily obtain information about the movie regardless of the title. Thus, an informative title affects consumer choice for under-promoted movies, but it is unlikely to affect their choice for movies accompanied by heavy promotion. The positive impact of an informative title is expected to decrease as the level of prerelease promotion increases.

Nevertheless, the problem of not always being able to directly measure the promotional expenditure of movies must be considered. Marketing budgets for blockbuster movies are often disclosed publicly (e.g., www.the-numbers.com); however, the budgets for under-promoted movies are often not disclosed at all. Because of this difficulty, this study uses the volume of media exposure as a proxy for the promotional budget for a movie. Media exposure is expected to be positively related to the promotional budget because media exposure increases market awareness, and market awareness is used as a proxy for the promotional budget, as discussed in the literature (e.g., Moon, Bayus, Yi, & Kim, 2015). Thus, we propose that the likelihood that movies are chosen increases when their titles are informative, but the effect will decrease as the prerelease media coverage of the movie increases.

H2. The impact of informative movie titles on box office revenue decreases as prerelease media coverage of the movies increases.

3. Methodology

3.1. Data collection

At 1.5 billion dollars, South Korea has the sixth-largest movie market in the world (MPAA, 2017). South Korea's well-developed Internet infrastructure enables the rapid diffusion of e-WOM through social network services and YouTube. Owing to such proliferation of e-WOM, many blockbuster movies, such as *After Earth*, *Iron Man 3*, *Oblivion*, *Life of Pi*, and *Cloud Atlas* were released in South Korea earlier than in the United States (Park, 2013). Therefore, this study analyzed the South Korean market to strongly reflect the effect of e-WOM.

All movies released in South Korea between 2012 and 2016 were considered in this study. The movies that were still showing at the end of each week were tallied to conduct weekly regression (e.g., Liu, 2006; Moon, Bergey, & Iacobucci, 2010), which resulted in 2734 movies that had revenue data for at least the first week of opening. We collected data for these movies for a maximum of 8 weeks after opening, if they were playing until then. Then we collected 8623 critic reviews from

Table 3
Control variables and their definitions.

Variable name	Definition
<i>lnScreen</i>	The natural log value of the number of screens on the premiere day.
<i>Sequel</i>	Dummy variable whose value is one for a sequel movie.
<i>lnVolume</i>	The natural log of the number of viewer reviews for movies added by one at Daum Movie (movie.daum.net), calculated weekly. For week 1, prerelease data were used.
<i>Valence</i>	Viewer's average review score (1–10) of movies at Daum Movie (movie.daum.net), calculated weekly. For week 1, prerelease data were used.
<i>CriticVolume</i> , <i>CriticValence</i>	Volume and valence (1–10) of critic reviews at Naver Movie (movie.naver.com).
<i>Korea</i> , <i>US</i> , <i>England</i> , <i>Japan</i> , <i>France</i>	The five countries of production that release the most movies in South Korea. Other countries are set as the baseline.
<i>Age12</i> , <i>Age15</i> , <i>Age18</i>	Movie content ratings. South Korean movie content ratings are for all ages, 12 years and older, 15 years and older, and 18 years and older. "For all ages" is set as the baseline.
<i>Drama</i> , <i>Action</i> , <i>Thriller</i> , <i>Children</i> , <i>Comedy</i> , <i>SF</i>	Genre. Followed the classification of Dellarocas et al. (2007). Genre coding is allowed to overlap, so no baseline was set. For example, <i>Captain America: Civil War</i> has three genres, Action, Thriller, and SF. The choice of genre followed the suggestions of the Korean Film Council.
<i>StarPower</i>	Sum of the audiences for all the movies starring the main actor of target movie during the last 3 years.
<i>Mon1–Mon11</i>	December is set as the baseline, and the movies released between January and November were dummy coded with <i>Mon1</i> to <i>Mon11</i> .

Naver Movie (movie.naver.com). Naver, which is similar to Google and Yahoo, is the largest portal site in South Korea, with a search market share of 77% in 2015 (Return on Now, 2015). For e-WOM, volume and valence data were collected from Daum Movie. Daum is the second-largest portal site in South Korea.² Title, genre, actors, release date, revenue, and the number of screens for each movie were obtained from the Korean Film Council (www.kofic.or.kr).

3.2. Dependent variable and independent variables

The independent and dependent variables used in this study are as follows:

lnRev: The natural log value of the box office revenue for each week is used as the dependent variable of this study.

lnMedia: The number of all news and magazine publications that mentioned the movie from the seventh day before opening to the day before opening was totaled, plus 1 (to define log extensively), and the natural log taken. This variable is the proxy for the pre-release promotional level of the movie.

Information: We define a dummy variable, *Information*, as the variable used to indicate an informative movie title. Providing access to information about the movie in the title creates an opportunity for consumers to infer the genre or storyline of the movie. A name with information has been defined as a suggestive brand name (Keller, Heckler, & Houston, 1998) in a brand context and as a descriptive title (Newmark, 1988) in a movie context. This study uses Newmark's (1988) method to determine whether a movie title is informative where Newmark (1988) classified movie titles into descriptive or allusive titles. On the one hand, a descriptive title conveys the theme of the movie and helps consumers to infer its genre or storyline. For example, the title *The Jungle Book* prompts consumers to assume that the movie is about a story related to a jungle. On the other hand, an allusive title refers literally or figuratively to the theme, but does not provide additional information. Therefore, the exact meaning of the title may be difficult to decipher and may only be revealed after watching the movie. For example, the title *Forrest Gump* suggests no information about the genre or

Table 4
Summary statistics of continuous variables.

Name	Mean	Std. Dev.	Min	Max
<i>lnRev</i>	17.77	2.91	10.46	24.65
<i>lnMedia</i>	2.78	1.92	0	8.35
<i>lnScreen</i>	5.56	1.97	1.95	9.49
<i>lnVolume</i>	1.32	1.28	0	7.08
<i>Valence</i>	5.59	4.09	0	10
<i>CriticVolume</i>	3.15	3.24	0	18
<i>CriticValence</i>	4.31	2.89	0	9.5
<i>StarPower</i>	1,113,008	2,547,947	0	2.94×10^7

storyline before watching the movie; however, consumers who have seen the movie can fully understand the meaning of the title. Newmark (1988) recommended generally using descriptive titles because they offer more information about the movie than allusive titles do. This study also claims that descriptive titles provide more information. In this study, *Information* is a dummy variable defined to be 1 if the title is informative (or descriptive, equivalently) or else 0. Three human coders categorized each movie title into informative or non-informative titles to measure *Information*. Two coders first viewed each movie title and decided whether it was informative or not. If they disagreed, the third coder weighed in to produce the result. The rate of agreement between the first two coders was 83.03%. Table 2 illustrates examples of how the variable *Information* was coded.

3.3. Control variables

Control variables, such as the number of screens, volume and valence of e-WOM, volume and valence of critic reviews, production country, viewing (movie content) ratings,³ genre, star power, and release month were also included in the model.⁴ The definitions of the control variables are presented in Table 3.

³ These ratings are similar to MPAA ratings in the US.

⁴ Social media has been identified as one of the major variables explaining market outcomes (Dellarocas et al., 2007). Thus, this study considered the number of blog entries on the movies as one of the control variables. We randomly selected 300 movies, which is about 10% of the total sample, and tracked entries made on these movies. As a result, the average number of postings was 10.17, and the number of postings was 0 for 115 movies, meaning that the majority of movies do not get social media exposure and the data were very skewed. Thus, social media was not included in the final model used in the study.

² Naver Movie does not disclose user rating data that exceeds 10,000 ratings, so we could not track weekly ratings. Thus, the data were obtained from an alternative source, Daum. The distribution of demographic information of Daum's users is expected to be similar to that of Naver's users. In fact, the average ratings of films in Daum used in this study were similar to those in Naver.

Table 5
Summary statistics of categorical variables.

Type	Proportion
Information	Information (38.04%), non-Information (61.96%)
Sequel	Sequel (6.84%), non-Sequel (93.16%)
Country	Korea (29.33%), US (28.24%), England (5.08%), Japan (10.13%), France (6.55%)
Viewing ratings	Age12 (18.47%), Age15 (33.87%), Age18 (28.86%)
Genre (multiple genres possible)	Drama (58.41%), Action (25.42%), Thriller (20.81%), Children (17.26%), Comedy (16.06%), SF (10.39%)
Month	Mon1 (7.46%), Mon2 (7.68%), Mon3 (8.08%), Mon4 (7.94%), Mon5 (8.19%), Mon6 (7.43%), Mon7 (7.72%), Mon8 (8.60%), Mon9 (8.34%), Mon10 (10.10%), Mon11 (10.06%)

3.4. Descriptive statistics and model-free evidence

The summary statistics of the variables on the opening week are presented in Tables 4 and 5.

To observe model-free evidence, 2734 movie samples in the first week were divided into four groups according to the size of *lnMedia*, and the averages of the variables *Information*, *lnRev*, *lnScreen*, *lnVolume*, *Valence*, *CriticVolume*, *CriticValence*, and *StarPower* were examined for each group, where group 1 has the smallest size of *lnMedia*, and group 4 enjoys the most media exposure. The results are shown in Fig. 1.⁵

It can be seen that all the variables except *Information* increase as the group index increases (i.e., as the amount of media exposure increases). This indicates that the promotional level, which is measured by media exposure, is positively related to *StarPower*, revenue (*lnRev*), *lnScreen*, and third-party reviews (*Valence*, *CriticValence*). In contrast, *Information* decreases from group 1 to group 3 but increases in group 4. This suggests that the studios with low levels of promotional activity seem to consciously or unconsciously know that providing information in movie titles is important, and that studios with high levels of marketing activity do not seem concerned with how informative a movie title is. We also examined the data by excluding sequels, remakes, and movies from original sources, and the same results held. This implies that for movies with low levels of promotional activity, including information in the titles becomes more important for getting the word out to potential viewers.

3.5. Model

The following regression model tests the main hypotheses of this study:

$$\begin{aligned} \ln Rev_{i,t} = & \beta_0 + \beta_1 Information_{i,t} + \beta_2 \ln Media_{i,t} + \beta_3 Information_{i,t} \times \ln Media_{i,t} + \alpha_1 \ln Screen_{i,t} + \alpha_2 \ln Volume_{i,t} + \alpha_3 Valence_{i,t} + \alpha_4 CriticVolume_{i,t} + \alpha_5 \\ & CriticValence_{i,t} + \alpha_6 Korea_{i,t} + \alpha_7 US_{i,t} + \alpha_8 England_{i,t} + \alpha_9 Japan_{i,t} + \alpha_{10} France_{i,t} + \alpha_{11} Age12_{i,t} + \alpha_{12} Age15_{i,t} + \alpha_{13} Age18_{i,t} + \alpha_{14} \\ & Drama_{i,t} + \alpha_{15} Action_{i,t} + \alpha_{16} Thriller_{i,t} + \alpha_{17} Children_{i,t} + \alpha_{18} Comedy_{i,t} + \alpha_{19} SF_{i,t} + \alpha_{20} StarPower_{i,t} + \alpha_{21} Mon1_{i,t} + \alpha_{22} Mon2_{i,t} + \alpha_{23} \\ & Mon3_{i,t} + \alpha_{24} Mon4_{i,t} + \alpha_{25} Mon5_{i,t} + \alpha_{26} Mon6_{i,t} + \alpha_{27} Mon7_{i,t} + \alpha_{28} Mon8_{i,t} + \alpha_{29} Mon9_{i,t} + \alpha_{30} Mon10_{i,t} + \alpha_{31} Mon11_{i,t} + \epsilon_{i,t}, \end{aligned}$$

where *i* denotes the movies (*i* = 1, ..., 2734) and *t* denotes the weeks (*t* = 1, 2, ..., 8).

The model was estimated using ordinary least squares. The results of the analysis are presented in Table 6. The variance inflation factor to test multicollinearity was up to 4.75 (for *lnMedia*) and 2.12 on average; therefore, multicollinearity rarely exists.

4. Results

4.1. Main results

Table 6 outlines the parameter estimates for the regression model. In week 1, the coefficient of *Information* is estimated to have a significant value of 0.64, supporting H1 that informative movie titles positively affect box office performance. According to the results, when the volume of prerelease media is 0, the movies with informative titles have a 64% higher revenue than the movies with non-informative titles. However, the effect on revenue decreases by 6.7% as the volume of prerelease media increases by 1%, supporting H2.

This effect appears to weaken or disappear as *t* increases; that is, the impact of *lnMedia* on *lnRev* was not significant in weeks 2 and 3. In week 4, the impact was significant and positive (*p* < 0.05), but from week 5, the impact was not observed again. This result shows that the impact of prerelease marketing sharply decreases after the movie is released. The impact of *Information* decreased from weeks 1 through 5, but still existed and was significant. This indicates that the movie title has a significant impact on revenue for many periods after the release of the movie during its opening. However, the interaction between *lnMedia* and *Information* was significant during weeks 1, 3, 4, and 8; marginally significant during weeks 3 and 4; and insignificant during weeks 2, 5, 6, and 7. This implies that the impact of informative movie titles on box office revenue decreases as prerelease media exposure of the movies increases, but after the first week of release the influence of the interaction diminishes rapidly.

Therefore, the impact of *Information*, *lnMedia*, and their interactions is the strongest during the first week of opening, and rapidly decreases after that; that is, the impact of an informative title is the strongest

immediately after the movie opens. After the opening, movie revenue is more likely to be affected by other factors, such as e-WOM. Indeed, *lnVolume* was found to have a significantly positive impact on revenue until week 5.

4.2. Robustness test

4.2.1. Analysis without sequels, remakes, or movies from other original sources

When the movie in question is a sequel, it is more likely that the previous movie was successful and consumers are well aware of it. For example, *X-Men: Apocalypse* is based on previous movies in the *X-Men* franchise; therefore, consumers already know that the movie is about superheroes. Furthermore, if the movie has an original source, such as a

⁵ The complete table with the exact numbers are available upon request.

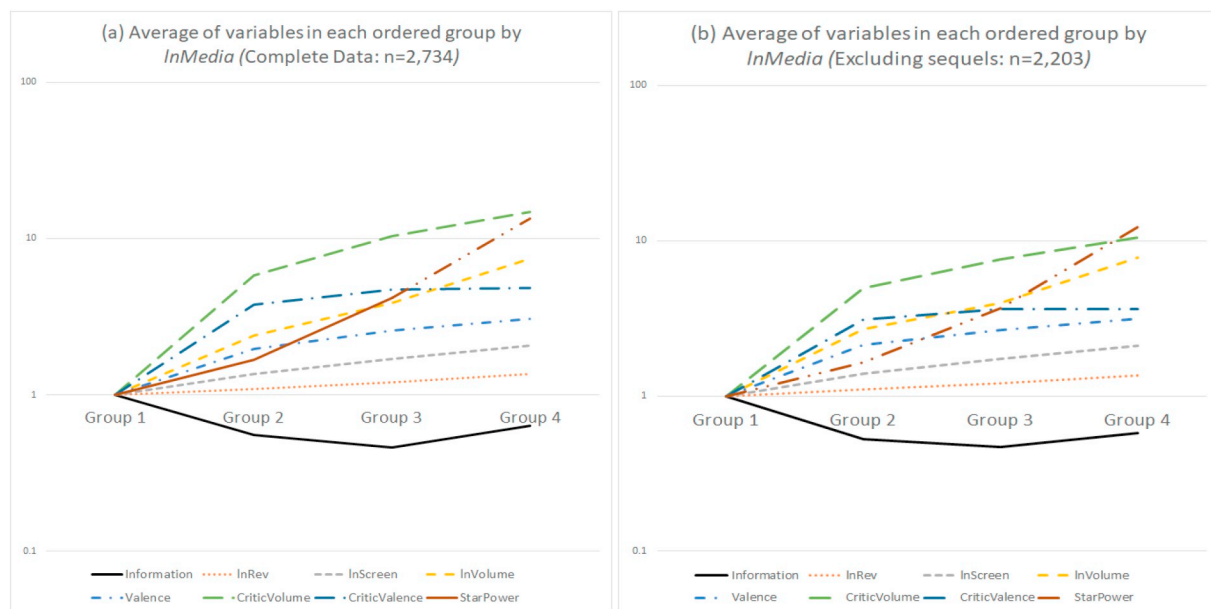


Fig. 1. The averages of the variables were examined for four groups, where group 1 has the smallest *lnMedia* and group 4 has the largest. (a) Complete data: $n = 2734$. (b) Data excluding sequels, remakes, and movies based on an original TV series, original novels, or comic books: $n = 2203$.

Note: The numbers used for the graph are relative values that are normalized by the values in group 1. The y-axis is in logarithmic scale.

Table 6

Main results with linear regression.

Week (t)	1	2	3	4	5	6	7	8
<i>lnMedia</i>	0.14***	0.028	0.024	0.055*	0.014	0.038	−0.026	0.015
<i>Information</i>	0.64***	0.35***	0.34***	0.40***	0.29*	0.23	−0.069	0.70*
<i>lnMedia*Information</i>	−0.067***	−0.031	−0.038 ⁺	−0.048 ⁺	−0.028	−0.0037	0.047	−0.13*
<i>lnScreen</i>	1.15***	1.23***	1.19***	1.19***	1.23***	1.28***	1.23***	1.27***
<i>Sequel</i>	0.13 ⁺	−0.016	−0.064	−0.26**	−0.13	−0.23 ⁺	0.14	0.39 ⁺
<i>lnVolume</i>	0.25***	0.26***	0.32***	0.24***	0.15***	0.048	0.052	0.024
<i>Valence</i>	−0.015**	0.00051	0.0026	0.0013	−0.0044	−0.0040	0.019	0.025
<i>CriticVolume</i>	0.083***	0.029**	0.019*	0.0088	0.017	0.011	0.15	0.021
<i>CriticValence</i>	−0.054***	−0.0095	−0.012	−0.018	−0.035*	−0.045*	−0.029	−0.049
<i>Korea</i>	−0.25***	−0.38***	−0.58***	−0.64***	−0.63***	−0.61***	−0.53**	−0.66***
<i>US</i>	0.20***	0.085	0.043	0.17*	0.025	0.081	0.025	−0.11
<i>England</i>	0.042	0.080	0.11	0.28*	0.31*	0.14	0.12	−0.27
<i>Japan</i>	−0.060	−0.026	−0.039	−0.031	−0.20	−0.10	−0.070	−0.028
<i>France</i>	0.16*	0.14 ⁺	0.098	0.20*	0.038	0.19	0.17	−0.044
<i>Age12</i>	−0.13*	−0.24***	−0.13*	−0.17*	−0.21*	−0.34**	−0.16	−0.25
<i>Age15</i>	−0.14*	−0.31***	−0.14*	−0.15*	−0.28**	−0.29**	−0.094	−0.13
<i>Age18</i>	−0.066	−0.26**	−0.18**	−0.22**	−0.34**	0.26*	−0.26	−0.11
<i>Drama</i>	−0.030	0.080 ⁺	0.020	0.060	0.16*	0.15	0.17	0.31 ⁺
<i>Action</i>	−0.0037	−0.077	−0.23***	−0.14*	−0.12	0.046	0.20	0.093
<i>Thriller</i>	0.048	−0.073	−0.13*	−0.099	0.00080	−0.044	−0.16	0.12
<i>Children</i>	0.13 ⁺	0.20*	0.36***	0.28**	0.085	−0.076	0.063	0.068
<i>Comedy</i>	−0.045	0.051	0.038	0.011	0.14 ⁺	0.014	−0.15	−0.13
<i>SF</i>	0.049	−0.060	−0.14*	−0.042	0.056	0.012	−0.16	0.16
<i>ActorPower</i>	4.40e-08***	3.27e-08***	1.83e-08**	2.64e-08***	1.16e-08	1.32e-08	7.58e-10	1.13e-08
<i>Mon1</i>	0.023	−0.0011	−0.019	0.13	0.27*	−0.056	−0.040	−0.11
<i>Mon2</i>	−0.092	−0.19*	−0.17*	−0.038	−0.16	−0.40*	−0.31	−0.36
<i>Mon3</i>	−0.29***	−0.34***	−0.18*	−0.12	−0.14	−0.18	−0.047	−0.0015
<i>Mon4</i>	−0.32***	−0.27**	−0.081	0.077	−0.011	−0.27 ⁺	−0.43 ⁺	−0.47
<i>Mon5</i>	−0.20*	−0.15 ⁺	−0.023	0.093	0.036	−0.059	0.072	−0.13
<i>Mon6</i>	−0.074	−0.074	−0.035	0.022	−0.077	−0.18	−0.090	0.079
<i>Mon7</i>	−0.033	0.088	0.17*	0.34**	0.27*	0.054	0.053	−0.10
<i>Mon8</i>	0.0075	0.098	0.12	0.24*	0.12	−0.13	−0.18	−0.063
<i>Mon9</i>	−0.15*	−0.12	−0.15 ⁺	−0.044	−0.12	−0.33*	−0.17	−0.054
<i>Mon10</i>	−0.20*	−0.19*	−0.096	0.044	−0.016	−0.21	0.22	0.33
<i>Mon11</i>	−0.29***	−0.19*	−0.026	0.095	0.18	0.044	0.046	−0.036
Constant	10.72***	10.21***	10.50***	10.65***	11.15***	11.40***	11.48***	11.26***
Observations	2734	2194	1580	1132	784	516	347	245
R ²	0.91	0.93	0.94	0.93	0.92	0.92	0.91	0.91

⁺ $p < 0.1$.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

Table 7

Results of subsample analysis after excluding movies that are sequels, remakes, or from other original sources.

Week (t)	1	2	3	4	5	6	7	8
<i>lnMedia</i>	0.14***	0.037 ⁺	0.032 ⁺	0.059*	0.028	0.025	−0.063	−0.011
<i>Information</i>	0.70***	0.39***	0.23*	0.35*	0.31 ⁺	0.35 ⁺	0.12	0.65*
<i>lnMedia*Information</i>	−0.098***	−0.039 ⁺	−0.0049	−0.026	−0.022	−0.036	−0.0039	−0.13 ⁺
<i>lnScreen</i>	1.12***	1.22***	1.21***	1.22***	1.24***	1.29***	1.26***	1.28***
<i>lnVolume</i>	0.29***	0.27***	0.32***	0.23***	0.15***	0.079 ⁺	0.083	0.0050
<i>Valence</i>	−0.022***	−0.000048	0.0023	0.0040	−0.0075	−0.014	0.0077	0.027
<i>CriticVolume</i>	0.080***	0.027**	0.018 ⁺	0.0055	0.015	0.011	0.011	0.017
<i>CriticValence</i>	−0.045***	−0.015	−0.010	−0.011	−0.036*	−0.062*	−0.019	−0.025
Results for other control variables are omitted								
Constant	10.83***	10.32***	10.51***	10.57***	11.15***	11.48***	11.35***	10.96***
Observations	2203	1750	1240	878	607	410	276	196
R ²	0.90	0.92	0.94	0.93	0.92	0.93	0.92	0.92

⁺ p < 0.1.

* p < 0.05.

** p < 0.01.

*** p < 0.001.

Table 8

Results of analysis with ultra-low budget subsamples.

Week (t)	1	2	3	4	5	6	7	8
<i>lnMedia</i>	0.092	0.21*	0.13	0.17 ⁺	0.080	0.043	0.021	0.051
<i>Information</i>	0.95***	0.69**	0.54	0.45	0.52	−0.18	0.54	1.21
<i>lnMedia*Information</i>	−0.061	−0.31*	−0.13	−0.047	−0.15	0.11	−0.10	−0.26
<i>lnScreen</i>	0.18	0.40	0.24	0.70*	0.41	0.61 ⁺	0.87*	0.96 ⁺
<i>lnVolume</i>	−0.31	0.16	0.40***	0.25 ⁺	−0.072	0.055	−0.13	−0.19
<i>Valence</i>	0.018	−0.019	−0.0028	−0.0047	0.028	−0.0093	0.025	0.022
<i>CriticVolume</i>	0.20	0.078	0.011	0.0077	0.041	−0.037	0.032	0.064
<i>CriticValence</i>	0.015	0.045	0.042	−0.0070	−0.083 ⁺	0.0046	0.021	−0.12
Results for other control variables are omitted								
Constant	13.84***	12.42***	12.51***	10.72***	13.57***	12.78***	12.22***	13.24***
Observations	262	235	184	154	154	125	91	73
R ²	0.27	0.29	0.43	0.44	0.36	0.38	0.40	0.40

⁺ p < 0.1.

* p < 0.05.

** p < 0.01.

*** p < 0.001.

book, comics, or TV program, there may be potential viewers who already know about its content and these people are no longer affected by its title. Therefore, we ran the analysis with a subsample that excluded all movies that are sequels, remakes, based on an original TV series broadcast in South Korea, based on original novels published in South Korea that are at least 6 months old, or based on a comic book. The results are shown in Table 7.⁶

The analysis shows that for week 1, the revenue increases by 71% on average when a movie title is informative. However, this effect decreases by 9.8% as the volume of prerelease media increases by 1%. The overall result is consistent with the results of the main model, which deals with the complete data, and the effect size is bigger than the main result. This indicates that the title effect appears to be stronger when movies that may already be known to potential viewers were excluded.

4.2.2. Analysis of ultralow-budget movies

Ultralow-budget movies lack accessibility to viewers because of their low or nonexistent promotional budgets. In such cases, revenue is therefore more likely to be affected by how effectively the movie's title conveys information. In general, because the production budget is

highly associated with the expected revenue, the production budget and the release scale (the number of opening-week screens) can be expected to have a positive relationship. Therefore, this subsection describes how we used the number of opening-week screens as a proxy for the production budget and conducted a subsample analysis by selecting ultralow-budget movies with the lowest 10% opening-week screens.⁷ Indeed, for 41 movies (mainly blockbusters) that disclosed their production budget, the correlation between the production budget and the log of the number of opening-week screens was 0.68 (p < 0.001). This implies that the variables have a relatively high correlation and change in step. The results of the subsample analysis are shown in Table 8.

The analysis shows that the revenue increases by 95% when the movie title is informative. The coefficient of *lnMedia* is bigger than the same coefficient in the main model, which indicates that the effect of the title on revenue is bigger for ultralow-budget movies. In week 1, there was no longer an interaction effect of *lnMedia*, but the effect was found in week 2, partially supporting H2.

Meanwhile, media exposure, e-WOM volume, and the number of screens were no longer significant in the first week. The result shows that it is particularly important for ultralow-budget movies to ensure

⁶ Due to space restrictions, we present only parameter estimates for the main variables in the robustness check and post-hoc analyses. The complete results are available upon request.

⁷ Another subsample analysis was performed by only choosing “blockbuster” movies, which accounted for the top 10% of the first week of screening. The results no longer supported the research hypotheses.

Table 9
General findings of this study.

Samples	General finding
Main result	- In week 1, the coefficient of <i>Information</i> is estimated to have a significant value of 0.64, supporting H1. This effect exists for later periods. - The effect decreases by 6.7% as the volume of prerelease media increases by 1%, supporting H2. This effect appears to weaken or disappear as <i>t</i> increases. - Overall, the impact of <i>Information</i>, <i>lnMedia</i>, and their interactions is the strongest during the first week of opening and rapidly decreases after that. - After the opening, movie revenue is more likely to be affected by other factors, such as e-WOM. Indeed, <i>lnVolume</i> was found to have a significantly positive impact on revenue until week 5.
Subsample excluding sequels, remakes, and movies from original sources	- During week 1, the revenue increases by 70% on average when a movie title is informative, but this effect decreases by 9.8% as the volume of prerelease media increases by 1%. - The overall result is consistent with the main model.
Subsample with ultralow-budget movies	- The analysis shows that the revenue increases by 95% on average when the movie title is informative. The coefficient is higher than the coefficient of the main model, which indicates that title affects the revenue more for ultralow-budget movies. - In week 1, there was no interaction effect by <i>lnMedia</i> anymore, but the effect was found in week 2, partially supporting H2.
Subsample with blockbuster movies	- H1 and H2 are no longer supported.

that the movie title is informative so that the viewer becomes aware of these movies and puts them in their consideration set.

5. Discussion and conclusion

5.1. Discussion

Table 9 summarizes the general findings of the analyses for the complete sample and the subsamples.

In the main analysis and the subsample analysis excluding sequels, the effect of informative movie titles and the interaction with *lnMedia* was strongest in the first week, and then the effect decreased rapidly in the subsequent weeks. However, for ultralow-budget movies, the effect of the informative movie title appeared to be larger, but this effect was not found for blockbuster movies. Therefore, the results of this study consistently emphasize that informative movie titles become more important in movies with low promotional budgets.

Finally, for illustration purposes, we conducted a post-hoc analysis to investigate how the informative movie title for the four groups divided by the size of media exposure (in Fig. 1) affected their box office revenue. The results of the subsample regression for each group are shown in Table 10.

The coefficient of *Information* was significant with only in group 1 and group 2 (which were the two groups that had the lowest volume of media exposure) among the four groups. This again confirms the hypothesis that an informative movie title is important when the

prerelease promotional level is low. When all other variables are held constant, the revenue of movies that have informative titles is 72% higher in group 1 and 105% higher in group 2 than for movies that have a non-informative title. In contrast, no significant effect was found in groups 3 and 4. Thus, all analyses consistently show that having an informative title is important for a movie, especially for those with fewer resources for promotion.

5.2. Theoretical implications

Under-promoted movies, which are usually produced outside of major film studios and lack sufficient production budgets, add diversity to the industry and fulfill various viewers' tastes and preferences. This study emphasized the importance of a movie's title, that is, an informative title has a positive impact on box office performance, especially for under-promoted movies, in the early weeks after opening. We measured prerelease promotional activity by the volume of prerelease media exposure and ran an ordinary least squares regression and saw a significant effect of informative movie titles. However, this impact decreases as prerelease promotional activities increase, and as time passes after the release of the movie.

Academically, this is the first study to examine the impact of an informative name or title on market outcomes in the entertainment industry. The results were robust for different settings. Movies are experience goods where viewers have uncertainty of quality before actually consuming them; thus, they try to collect various information about them to reduce the uncertainty. A movie's title is one source of information that potential consumers can be easily exposed to, and it can affect their behavior. The mechanism we demonstrated in the study not only contributes to the understanding of movies, but it can also be applied to the consumption of other goods in the entertainment industry, as well as to the literature on brand names in general.

5.3. Managerial implications

From a managerial perspective, because an informative movie title positively affects the revenue of under-promoted movies, the movie studios should create titles that reveal key information about the movie. The title effect was found to be stronger for ultralow-budget movies where e-WOM appeared to be insignificant. This result does not imply that e-WOM is not helpful. Rather, even though e-WOM has been considered as an effective marketing tool where resources are relatively scarce, ultralow-budget movies cannot rely on consumer e-WOM, as it is difficult for them to create enough e-WOM in the first place to create publicity. Therefore, producers of ultralow-budget movies should pay particular attention to naming the movie. Further, marketers in relevant

Table 10
Results of post-hoc analysis.

Group	1	2	3	4
<i>lnMedia</i>	0.065	0.25 ⁺	0.18	0.074 ⁺
<i>Information</i>	0.72***	1.05 ⁺	0.31	0.32
<i>lnMedia</i> * <i>Information</i>	0.045	-0.35	-0.049	-0.025
<i>lnScreen</i>	0.89***	1.05***	1.11***	1.42***
<i>lnVolume</i>	0.46**	0.12*	0.19***	0.21***
<i>Valence</i>	-0.051 ⁺	0.014	-0.0095	0.022 ⁺
<i>CriticVolume</i>	0.035	0.082***	0.034 ⁺	0.048***
<i>CriticValence</i>	-0.011	-0.053**	-0.026	-0.0028
Results for other control variables are omitted				
Constant	11.57***	10.98***	11.22***	8.53***
Observations	544	526	574	559
R ²	0.59	0.82	0.87	0.94

⁺ *p* < 0.1.

* *p* < 0.05.

** *p* < 0.01.

*** *p* < 0.001.

industries, such as the book, music, and TV industries, should give considerable attention to the information conveyed by their brands' or products' names.

5.4. Limitations and future research

However, this study is not without limitations, and hence we provide some directions for future research. First, although the study shows that it can be beneficial to reveal information about a movie in its title, adverse effects were not considered. A characteristic of a good brand name may be memorability, or the ease of remembering a name. Is there a tradeoff between the amount of information and memorability? For example, does a 10-line title that describes all the stories in the movie help increase revenue? Brand memorability is one of the most important factors in brand names (Robertson, 1989) and is a necessary element in information processing (Aaker, 1991). From this perspective, a moderately long title that is both easy to remember and informative is expected to be helpful. However, this study does not address this tradeoff. In the future, a study that involves a more precise consideration of these effects should be undertaken.

Second, this study did not use data from online social media platforms due to skewness in these data. However, it is possible that the interaction between the volume of social media and informative movie titles may have a positive impact on box office revenue, as informative movie titles can be shared more often among consumers on online social media platforms. We leave this subject for future research.

Third, the study domain can be extended to other countries. Essentially the same effect is expected for countries other than South Korea because the mechanism of the informative title effect is associated with a reduction in consumers' search costs and perceived risk. However, cross-cultural differences possibly moderate the result. Therefore, future research can test whether the results are consistent across different countries. If a different effect is found, it would be an interesting future topic to explore how the cross-cultural components moderate the effect of an informative title. In sum, we hope that this research may be a starting point for other fruitful research in the naming of products in the entertainment industry.

Declaration of Competing Interest

None.

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